CurriculumAI

Software Verification Test Cases

Test specification and execution summary | 28 October 2026

1. Purpose and scope

This document records 39 selected software verification cases for the CurriculumAI application. The cases cover unit/service, integration, functional, error/edge-case, browser/interface and full-system testing. They correspond to the case identifiers used in Chapter 5 of the dissertation.

The recorded evidence supports 39 passing selected cases across two contributing executions on 28 October 2026. One run completed the 36 controlled cases but subsequently encountered a live-setup failure; a separate run completed S01-S03 with live OpenAI integration. Earlier failed attempts were retained in the execution review. These results do not represent a single uninterrupted 39-case execution.

Testing used synthetic Scala questions, documents and accounts. Backend tests used disposable PostgreSQL/pgvector and isolated storage, with external services simulated as needed. Browser cases used Playwright with simulated API responses. The three full-system cases used actual React, Django, PostgreSQL/pgvector, the ingestion worker and live OpenAI integration in a disposable environment. These tests verify defined software workflows, not the educational accuracy of generated answers or student usability, which are evaluated separately in Chapter 6.

2. Results summary

Category	Case identifiers	Passed	Failed
Unit/service	U01-U06	6	0
Integration	I01-I08	8	0
Functional	FT01-FT06	6	0
Error and edge-case	E01-E08	8	0
Browser/interface	B01-B08	8	0
Full-system	S01-S03	3	0
Total		**39**	**0**

Table 1. Consolidated results for the 39 selected verification cases.

3. Test environment and evidence

Test environment. A disposable PostgreSQL/pgvector database, temporary media storage, synthetic test accounts and controlled third-party responses were used for the isolated checks.

Live workflows. S03, S01 and S02 passed in run v39-20261028t075257z-09a3dd using the disposable application and genuine OpenAI requests. Historical logs, system results and provider accounting are preserved in Git commit df8a31d.

Reproducibility. The selected tests and runner instructions are under test/ in the CurriculumAI repository. The original execution reports are preserved in Git commit df8a31df9c1acc146ab3a73b0bb969e75cb02557. Current report folders may contain later executions.

Reported additional checks. Existing regression suites also passed: 780 backend tests, 31 frontend unit tests and 160 browser tests. Django checks, migration checks, lint and the production frontend build passed. A non-failing bundle-size warning remained.

Evidence and test sources are organised in the following repository locations:

`test/thesis_verification/cases.json` - case definitions and requirement mapping

`test/thesis_verification/` - selected tests by category, plus shared fixtures and runner scripts

`test/reports/` - historical reports preserved in Git commit df8a31d; current report folders may contain later runs

`test/README.md` - instructions for reproducing the test runs

4. Individual test cases

Each case lists its requirements, procedure, expected outcome, recorded status and source location. PASS refers to the historical consolidated evidence across two contributing executions on 28 October 2026, not to a fresh execution of the later Docker-only launcher.

4.1 Unit and service tests

These cases exercise extraction, validation and decision rules using controlled inputs. They do not evaluate educational answer quality.

U01 - Document text extraction

Requirement(s): FR08, NFR06

Procedure: Provide a short supported PDF or Office document with identifiable text and source positions; run the applicable extractor; inspect extracted content and locator fields.

Expected outcome: Extracted text is preserved in reading order with usable source-location metadata.

Recorded result: PASS - historical consolidated evidence (28 October 2026)

Test source

`test/thesis_verification/unit_test/U01/`

U02 - Location-aware text chunking

Requirement(s): FR08, NFR06

Procedure: Pass extracted programming material to the chunking service; inspect chunk order, configured 1,200-character windows, overlap and associated locators.

Expected outcome: Chunks follow the configured boundaries and overlap while retaining source identity and ordering.

Recorded result: PASS - historical consolidated evidence (28 October 2026)

Test source

`test/thesis_verification/unit_test/U02/`

U03 - Citation validation rules

Requirement(s): FR03, NFR02

Procedure: Supply controlled generated responses containing unknown, malformed, missing or duplicated citation entries, together with reconstructed or non-contiguous quotations.

Expected outcome: Unknown, malformed, missing or duplicated citation entries and reconstructed/non-contiguous quotations are rejected.

Recorded result: PASS - historical consolidated evidence (28 October 2026)

Test source

`test/thesis_verification/unit_test/U03/`

U04 - Whole-question evidence assessment

Requirement(s): FR02, FR03, FR04

Procedure: Supply controlled low-similarity but sufficient evidence and high-similarity but incomplete evidence, together with simulated support assessments.

Expected outcome: Routing does not equate passage similarity with support for every substantive question part.

Recorded result: PASS - historical consolidated evidence (28 October 2026)

Test source

`test/thesis_verification/unit_test/U04/`

U05 - Fallback reason classification

Requirement(s): FR04-FR05

Procedure: Exercise eligible insufficiency or validation reasons and protected authentication, provider and integrity errors.

Expected outcome: Only approved reasons can initiate Tier 2; protected conditions return controlled errors.

Recorded result: PASS - historical consolidated evidence (28 October 2026)

Test source

`test/thesis_verification/unit_test/U05/`

U06 - Request-local generation-model selection

Requirement(s): FR11, NFR07

Procedure: Configure distinct Tier 1 and Tier 2 models, then test model-selection snapshots across generation and recovery operations.

Expected outcome: Each request uses its selected model consistently without unintended cross-request changes.

Recorded result: PASS - historical consolidated evidence (28 October 2026)

Test source

`test/thesis_verification/unit_test/U06/`

4.2 Integration tests

These cases examine connected backend services, persistence and real PostgreSQL/pgvector behaviour, with simulated external responses where appropriate.

I01 - PDF ingestion to retrieval

Requirement(s): FR02, FR08, NFR03

Procedure: Submit a synthetic PDF through the ingestion path; process extraction, chunking and embeddings; query the indexed source through retrieval.

Expected outcome: A successfully indexed document can provide retrievable evidence with source locations.

Recorded result: PASS - historical consolidated evidence (28 October 2026)

Test source

`test/thesis_verification/integration_test/I01/`

I02 - Retrieval, generation and answer persistence

Requirement(s): FR01, FR03, FR07, FR08, NFR04

Procedure: Retrieve indexed passages, supply a controlled structurally valid Tier 1 generation response and inspect stored conversation and citation records.

Expected outcome: A validated response, encrypted messages and a citation snapshot are saved consistently.

Recorded result: PASS - historical consolidated evidence (28 October 2026)

Test source

`test/thesis_verification/integration_test/I02/`

I03 - Citation repair and exhausted repair

Requirement(s): FR03, FR05, NFR03

Procedure: Feed a response containing a non-empty incorrect quotation; permit one repair; repeat with a second invalid quotation.

Expected outcome: Successful repair stores one valid response; exhausted repair does not persist rejected Tier 1 content and follows the permitted recovery route.

Recorded result: PASS - historical consolidated evidence (28 October 2026)

Test source

`test/thesis_verification/integration_test/I03/`

I04 - Private conversation access and encryption

Requirement(s): FR07, NFR04

Procedure: Create a registered conversation; access it using its owner and another user; inspect stored private fields and decryption order.

Expected outcome: Private content is encrypted at rest and access is denied to non-owners before decryption.

Recorded result: PASS - historical consolidated evidence (28 October 2026)

Test source

`test/thesis_verification/integration_test/I04/`

I05 - Historical citations after source deletion

Requirement(s): FR03, FR09, NFR02

Procedure: Create a sourced answer, delete its knowledge source, reload citation records and decrypt the stored assistant message.

Expected outcome: Stored citation snapshots preserve the historical source title and location after the source record is deleted.

Recorded result: PASS - historical consolidated evidence (28 October 2026)

Test source

`test/thesis_verification/integration_test/I05/`

I06 - Independent model settings and restart persistence

Requirement(s): FR11, NFR07

Procedure: Update Tier 1 and Tier 2 selections, inspect audit entries, restart the Django process in the disposable setup and read configuration again.

Expected outcome: Tier selections remain independent and auditable, survive stopping and restarting an isolated Django server, and roll back if audit insertion fails.

Recorded result: PASS - historical consolidated evidence (28 October 2026)

Test source

`test/thesis_verification/integration_test/I06/`

I07 - Concurrent uploads and worker exclusion

Requirement(s): FR08, NFR03

Procedure: Submit matching document binaries concurrently and attempt overlapping ingestion work; inspect database reservation and progress stages.

Expected outcome: Duplicate reservation is prevented; processing stages remain consistent and conflicting worker execution is excluded.

Recorded result: PASS - historical consolidated evidence (28 October 2026)

Test source

`test/thesis_verification/integration_test/I07/`

I08 - Real pgvector cosine ranking

Requirement(s): FR02, NFR03

Procedure: Insert synthetic relevant and irrelevant vectors into the pgvector-enabled test database and run retrieval ranking.

Expected outcome: The relevant record ranks ahead of irrelevant records under real cosine-distance ordering.

Recorded result: PASS - historical consolidated evidence (28 October 2026)

Test source

`test/thesis_verification/integration_test/I08/`

4.3 Functional tests

These cases verify expected student and administrator application operations.

FT01 - Student account operations

Requirement(s): FR06, NFR04

Procedure: Exercise registration, verification, login/logout, account recovery, supported email/password changes, and account-deletion scheduling, repeated scheduling and undo using synthetic accounts.

Expected outcome: Valid operations succeed; invalid or unauthorised requests are rejected; deletion scheduling and undo are consistent. Permanent deletion completion is not established by this case.

Recorded result: PASS - historical consolidated evidence (28 October 2026)

Test source

`test/thesis_verification/functional_test/FT01/`

FT02 - Guest question answering

Requirement(s): FR01, NFR04

Procedure: Submit a conceptual Scala question without signing in and inspect the generated response and storage state.

Expected outcome: A guest can receive a response while no registered conversation history is created.

Recorded result: PASS - historical consolidated evidence (28 October 2026)

Test source

`test/thesis_verification/functional_test/FT02/`

FT03 - Registered conversation management

Requirement(s): FR01, FR07, NFR04

Procedure: Create and continue a conversation, reopen it, rename it and delete it using a registered test student.

Expected outcome: Only the owner can manage the conversation; updates and deletion behave as specified.

Recorded result: PASS - historical consolidated evidence (28 October 2026)

Test source

`test/thesis_verification/functional_test/FT03/`

FT04 - Tier 1 response persistence and citation identity

Requirement(s): FR03, NFR02

Procedure: Submit a course-supported question with controlled Tier 1 generation; inspect encrypted message and citation records and the original marker identity.

Expected outcome: A validated Tier 1 response and its citations persist correctly, with encrypted messages and preserved citation marker identity. Rendering of tier labels is tested separately.

Recorded result: PASS - historical consolidated evidence (28 October 2026)

Test source

`test/thesis_verification/functional_test/FT04/`

FT05 - Tier 2 routing, recovery and persistence

Requirement(s): FR04, FR05, NFR02

Procedure: Exercise backend empty-corpus fallback, insufficient-support routing and malformed-response recovery; inspect reason codes, persisted output and course-citation records.

Expected outcome: Eligible requests reach Tier 2 with appropriate reasons; accepted output is stored without course citations, and rejected Tier 1 responses are not saved. Interface presentation is tested separately.

Recorded result: PASS - historical consolidated evidence (28 October 2026)

Test source

`test/thesis_verification/functional_test/FT05/`

FT06 - Administrator account management

Requirement(s): FR12, NFR04

Procedure: Test administrator login, password/email changes and account-access protections. Check the shared logout endpoint using a student session; do not treat this as a separately exercised administrator logout.

Expected outcome: Administrator login and account-security changes meet the enforced contracts; the shared logout endpoint clears the exercised student session; unauthorised access is denied.

Recorded result: PASS - historical consolidated evidence (28 October 2026)

Test source

`test/thesis_verification/functional_test/FT06/;`
`test/thesis_verification/functional_test/FT01/test_auth.py:41 (shared logout)`

4.4 Error and edge-case tests

These cases examine invalid input, boundary conditions, access restrictions and controlled recovery.

E01 - Invalid or oversized uploads

Requirement(s): FR08, NFR03, NFR04

Procedure: Check declared file sizes against the 200,000,000-byte application boundary; use a reduced 32-byte test limit for actual stream rejection, alongside invalid signatures and package structures.

Expected outcome: Unsafe or invalid files are rejected. Size validation is tested at the declared boundary and with a small actual stream, not by uploading a 200 MB fixture.

Recorded result: PASS - historical consolidated evidence (28 October 2026)

Test source

`test/thesis_verification/edge_case_test/E01/`

E02 - Unsafe website destinations and redirects

Requirement(s): FR08, NFR04

Procedure: Submit controlled forbidden URL targets and redirects using isolated network fixtures rather than live private addresses.

Expected outcome: Unsafe source destinations or redirect targets are rejected.

Recorded result: PASS - historical consolidated evidence (28 October 2026)

Test source

`test/thesis_verification/edge_case_test/E02/`

E03 - Duplicate upload and ingestion failure isolation

Requirement(s): FR08, NFR03

Procedure: Submit duplicate binaries, simulate an all-zero embedding followed by another valid queued source, and retry failed ingestion with re-extraction and replacement of partial chunks.

Expected outcome: Duplicate reservation is enforced; an invalid source fails without stopping subsequent work; retry re-extracts content and replaces partial chunks safely.

Recorded result: PASS - historical consolidated evidence (28 October 2026)

Test source

`test/thesis_verification/edge_case_test/E03/`

E04 - Empty or partially supported corpus

Requirement(s): FR02, FR04-FR05

Procedure: Run controlled empty-corpus and partially supported question scenarios; inspect routing decisions.

Expected outcome: Fallback or controlled handling matches the specified evidence and reason conditions.

Recorded result: PASS - historical consolidated evidence (28 October 2026)

Test source

```
test/thesis_verification/edge_case_test/E04/;
test/thesis_verification/unit_test/U04/test_hybrid_routing.py:42 (shared partial-support
assertion)
```

E05 - Malformed or fabricated citations

Requirement(s): FR03, FR05, NFR02

Procedure: Supply fabricated or duplicate source markers and quotations altered from retrieved text.

Expected outcome: Citation validation rejects invalid attribution rather than silently presenting it as verified.

Recorded result: PASS - historical consolidated evidence (28 October 2026)

Test source

```
test/thesis_verification/edge_case_test/E05/
```

E06 - Protected provider and persistence failures

Requirement(s): FR05, NFR04

Procedure: Simulate provider outage or refusal, permission failure, encryption failure and database persistence error.

Expected outcome: Protected failures produce controlled errors and never invoke inappropriate Tier 2 recovery.

Recorded result: PASS - historical consolidated evidence (28 October 2026)

Test source

```
test/thesis_verification/edge_case_test/E06/;
test/thesis_verification/unit_test/U05/test_technical_recovery.py:8 (shared protected-failure
assertion)
```

E07 - Ownership, role and CSRF enforcement

Requirement(s): FR07, FR12, NFR04

Procedure: Test cross-user conversation requests, administrator, superuser and disabled-account attempts against private conversation APIs, and invalid CSRF proof. Administrator security-endpoint restrictions are checked in FT06.

Expected outcome: Conversation ownership and account-role restrictions are enforced before sensitive processing; CSRF-protected operations reject invalid requests.

Recorded result: PASS - historical consolidated evidence (28 October 2026)

Test source

```
test/thesis_verification/edge_case_test/E07/;
test/thesis_verification/functional_test/FT03/test_storage.py:27 (shared ownership assertion)
```

E08 - Stale source during final verification

Requirement(s): FR03, FR05, FR09, NFR03

Procedure: Invalidate a retrieved source before the final acceptance/persistence step and inspect rejected answer handling.

Expected outcome: Stale evidence is rejected, rejected Tier 1 output is not saved and only eligible recovery is attempted.

Recorded result: PASS - historical consolidated evidence (28 October 2026)

Test source

```
test/thesis_verification/edge_case_test/E08/
```

4.5 Browser and interface tests

Playwright runs a real browser against simulated API responses. The browser cases do not establish live backend networking.

B01 - Student registration and login screens

Requirement(s): FR06, NFR01

Procedure: Navigate registration, verification and login screens using a synthetic identity; inspect feedback and transitions.

Expected outcome: Expected account states and navigation render correctly.

Recorded result: PASS - historical consolidated evidence (28 October 2026)

Test source

`test/thesis_verification/browser_test/B_cases.spec.js:38`

B02 - Guest chat interaction

Requirement(s): FR01, NFR01, NFR02

Procedure: Enter a guest question; submit it; inspect loading state, response text and origin label.

Expected outcome: The guest workflow renders the submitted question and returned response without a logged-in account.

Recorded result: PASS - historical consolidated evidence (28 October 2026)

Test source

`test/thesis_verification/browser_test/B_cases.spec.js:52`

B03 - Conversation history interface

Requirement(s): FR07, NFR01

Procedure: In a simulated-browser session, reopen and rename a registered conversation, confirm deletion and verify that the deleted item is removed.

Expected outcome: History navigation and renaming work; confirmed deletion removes the conversation from the interface.

Recorded result: PASS - historical consolidated evidence (28 October 2026)

Test source

`test/thesis_verification/browser_test/B_cases.spec.js:71`

B04 - Expandable citations and deleted source

Requirement(s): FR03, FR09, NFR02

Procedure: Display a Tier 1 answer, expand its citations and show the deleted-source historical state.

Expected outcome: References expand with readable locators and retain explanatory snapshot details.

Recorded result: PASS - historical consolidated evidence (28 October 2026)

Test source

`test/thesis_verification/browser_test/B_cases.spec.js:99`

B05 - Tier 2 external response labelling

Requirement(s): FR04, FR05, NFR02

Procedure: Render controlled Tier 2 responses for all six fallback explanations and inspect their labels and reason-specific notices.

Expected outcome: All six reason explanations display a clear Tier 2 origin without misleading course citations.

Recorded result: PASS - historical consolidated evidence (28 October 2026)

Test source

`test/thesis_verification/browser_test/B_cases.spec.js:116`

B06 - Administrator source and usage views

Requirement(s): FR08, FR10, NFR01

Procedure: Navigate the administrator source list, inspect ingestion stages and check aggregate usage displays.

Expected outcome: Status, stage and usage information appears under the appropriate administrator interface.

Recorded result: PASS - historical consolidated evidence (28 October 2026)

Test source

`test/thesis_verification/browser_test/B_cases.spec.js:131`

B07 - Independent administrator model controls

Requirement(s): FR11, NFR07

Procedure: Change Tier 1 and Tier 2 selections through the intercepted administrator API; inspect the PATCH request and restore simulated API selections after reload.

Expected outcome: Tier controls submit the correct PATCH request and restore independent selections from simulated API data. Database persistence is verified separately in I06.

Recorded result: PASS - historical consolidated evidence (28 October 2026)

Test source

`test/thesis_verification/browser_test/B_cases.spec.js:146`

B08 - Registration across changed frontend deployment

Requirement(s): FR06, NFR01

Procedure: Simulate an unavailable deferred chat JavaScript bundle during the registration-to-dashboard transition.

Expected outcome: Registration navigation completes without requiring the unavailable asset.

Recorded result: PASS - historical consolidated evidence (28 October 2026)

Test source

`test/thesis_verification/browser_test/B_cases.spec.js:168`

4.6 Full-system tests

These three cases were exercised through the running disposable application and live OpenAI integration with synthetic content.

S01 - Complete student Tier 1 workflow

Requirement(s): FR01, FR03, FR07

Procedure: Open the real interface; log in using a synthetic student account; submit a supported Scala question; expand citations; reopen the saved chat.

Expected outcome: A labelled Tier 1 answer with citations is displayed, and the conversation can be reopened from saved history.

Recorded result: PASS - historical consolidated evidence (28 October 2026)

Test source

`test/thesis_verification/system_test/S_workflows.spec.js:71`

S02 - Complete Tier 2 fallback workflow

Requirement(s): FR04-FR05

Procedure: Submit a question unsupported by the synthetic course corpus through the running application; inspect response label and cited material.

Expected outcome: The system produces the eligible Tier 2 response with an explanatory label and no course citations.

Recorded result: PASS - historical consolidated evidence (28 October 2026)

Test source

`test/thesis_verification/system_test/S_workflows.spec.js:92`

S03 - Administrator ingestion-to-retrieval workflow

Requirement(s): FR02, FR03, FR08

Procedure: Log in as administrator; upload a synthetic Scala PDF; allow the independent worker to index it; ask a question dependent on the new source.

Expected outcome: The document reaches an indexed state and the newly available evidence is retrieved and cited.

Recorded result: PASS - historical consolidated evidence (28 October 2026)

Test source

`test/thesis_verification/system_test/S_workflows.spec.js:60`

5. Interpretation and limitations

All 39 selected cases have recorded passing evidence across two contributing executions, not one uninterrupted run. The first run completed 36 controlled cases but later failed during live setup; a separate live run passed S01-S03. Subsequent isolated reruns may show S01-S03 as blocked without fresh paid-provider approval, without negating the historical passes.

The verification covers specified functional behaviour, data handling, failure boundaries and integration. Tests using controlled or simulated AI outputs do not establish the factual correctness, relevance or source faithfulness of generated explanations. The three live system cases demonstrate exercised integration using synthetic questions and documents, not general performance or usability with actual students.

Earlier attempts remain distinguishable: live setup initially failed because httpx was absent; a subsequent system run timed out during logout navigation; another returned HTTP 503 when the test budget guard rejected the actual schema name before generation. The final live retest passed S01-S03. These were test-environment, navigation and instrumentation issues, not demonstrated failures of the application logic.

Appendix A Supplement

CurriculumAI - Historical Test Execution Evidence Index

A. Purpose and evidence boundary

This index links the 39 selected cases in Appendix A to their archived results, test sources and recorded revisions. It documents verification of specified behaviours under the recorded conditions.

B. Historical execution summary

Date and time (UTC)	Run ID	Environment	Recorded outcomes	Primary evidence
28 Oct 2026 07:12:00 to 07:22:27	v39-20261028t071200z-c653d5	Isolated Django/PostgreSQL and Playwright with simulated APIs	36 PASS 3 BLOCKED during live setup (missing httpx)	backend-cases.jsonl browser.json live-system-migrations.log
28 Oct 2026 07:52:57 to 07:54:03	v39-20261028t075257z-09a3dd	Disposable application and live OpenAI	S01-S03 PASS 36 NOT_EXECUTED	system.json provider-usage.json
30 Oct 2026 10:52:41	docker-thesis-20261030t105226z-1fbe8b	Isolated Docker	6 PASS Unit/service	results/summary.json
30 Oct 2026 10:54:01	docker-thesis-20261030t105350z-c5552b	Isolated Docker	8 PASS Integration	results/summary.json
30 Oct 2026 10:59:43	docker-thesis-20261030t105932z-8affd2	Isolated Docker	6 PASS Functional	results/summary.json
30 Oct 2026 11:01:48	docker-thesis-20261030t110139z-44cb69	Isolated Docker	8 PASS Error/edge-case	results/summary.json
30 Oct 2026 11:04:21	docker-thesis-20261030t110412z-3b693d	Isolated Docker	8 PASS Browser/interface	results/summary.json
30 Oct 2026 11:05:06	docker-thesis-20261030t110454z-abee72	Isolated Docker with live OpenAI	3 PASS Full-system	results/summary.json

C. Coverage reconciliation

The 28 October controlled run contributes six unit/service, eight integration, six functional, eight error/edge-case and eight browser/interface passes. The 28 backend cases account for 87 check references. The live run contributes three full-system passes. The 39 unique outcomes span these two executions. The six 30 October runs provide separate later evidence.

Companion mapping

CurriculumAI_39_Case_Evidence_Map.csv

Open the CSV beside this PDF. It gives one row per case with its result locator, source locations and evidence hash.

D. Primary evidence and source traceability

Original base revision 129ee5ebfef7d527d2a99d006ccd9335266015ab

Git archive commit df8a31df9c1acc146ab3a73b0bb969e75cb02557

The two original run summaries preserve 317 per-file hashes before and after execution. Eight entries differ between the two run snapshots. The CSV points to the build context for the run that exercised each case. The companion audit lists the common source-image fingerprint recorded by the six later Docker runs.

In an authorised clone, use git show <archive-commit>:test/reports/<run-id>/results/<file>. The case manifest and relocation map are under test/thesis_verification/. The companion audit supplies exact retrieval commands, result hashes and repository links for use with verified access.

Unauthenticated GitHub requests for the repository and commit returned HTTP 404. An examiner can use authorised repository access or reviewed evidence copies submitted through the approved university channel.

E. Representative integrity verification - E06

The Tier 1 service writes the assistant message before attempting the citation insert in the same atomic transaction. E06 injects an IntegrityError at citation insertion and checks for STORAGE_FAILURE, an unchanged conversation timestamp, the original student message as the only remaining message, and zero citation rows. A separate routing regression checks that this storage failure does not invoke Tier 2.

Exact method test_citation_insert_failure_rolls_back_assistant_and_conversation_timestamp

Current source test/thesis_verification/edge_case_test/E06/test_generation.py:7

The named rollback test recorded PASS in the 28 October controlled run and the 30 October edge-case run. Its evidence covers the injected citation-insert failure within this transaction.

F. Limitations

Controlled provider outputs cannot establish educational answer correctness. Browser tests with simulated APIs cannot establish live backend integration. The three live workflows use synthetic inputs and offer limited evidence about broader performance. Historical passes apply to the tested builds and scenarios. Security, scalability and reliability require evidence beyond this selected suite.